

MAYILRAJ R *Software Engineer*

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Tuticorin, Tamil Nadu, India

SUMMARY

Final-year IT undergraduate with hands-on internship experience across full-stack (MERN, Django) and backend engineering, plus independent projects spanning IoT (ESP32 sensor pipelines) and applied AI (computer vision, speech). Comfortable owning features end-to-end — API design, database optimisation, and test coverage — and looking for a full-time Software Engineer role starting mid-2026.

EXPERIENCE

Software Development Engineer Intern, *Ibee Analytics*

Aug 2025 – Sep 2025

Hyderabad

- Built Python backend modules (OOP/OOAD) with pytest coverage above 90% across 5+ API endpoints, cutting post-release bug discovery by ~30%.
- Shipped 3 RESTful APIs in Node.js/Express following a modular MVC structure, with pre-release impact analysis for each sprint.
- Built React.js analytics dashboards with Chart.js for real-time KPI visualisation; tuned MongoDB schemas (compound indexes, projection), reducing average read latency by ~25%.
- Worked across 4 Agile sprints using Git branching and PR-based code review.

Web Developer Intern, *Hapus Infotech*

Jul 2025 – Aug 2025

Coimbatore

- Refactored a legacy Python/Django codebase using Factory and Strategy design patterns, cutting average page-load time by ~20% and duplicated code by 35%.
- Implemented JWT authentication and session management with input validation; rebuilt the UI in Bootstrap 5, raising Lighthouse mobile usability from 62 to 87.

EDUCATION

B. Tech, Information Technology — *University College of Engineering, Tiruchirappalli*

08/2023 - 05/2026

CGPA: 7.2 / 10

Diploma, Electronics & Communication Engineering — *Thevanesam Eruthaya Ammal Polytechnic College*

06/2020 – 04/2022

CGPA: 9.1 / 10

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, Embedded C, SQL

Backend / APIs: Node.js, Express.js, Flask, Django, REST API Design

Frontend: React.js, HTML5, CSS3, Bootstrap 5, Chart.js

Databases: MongoDB, MySQL, PostgreSQL

Tools & Practices: Git/GitHub, Postman, VS Code, Agile/Scrum, OOP/OOAD, CI/CD

IoT & Applied AI: ESP32, Arduino, MQTT, ThingSpeak, MediaPipe, OpenCV, PyTorch

PROJECTS

Predictive Marketing Dashboard — *Python, Flask, Facebook Prophet, Beautiful Soup*

- Built a modular Flask backend (Repository, Service Layer patterns) with 95%+ pytest coverage; used Facebook Prophet to forecast marketing KPIs (<8% MAPE) and automated SEO data collection via Beautiful Soup, cutting manual reporting by ~4 hrs/week.

Empower Her — Safe Route Navigation — *Android Studio, Google Maps SDK, Firebase, AI/ML*

- Built an Android safety app with an AI-based threat detection module, BLE smart-ring SOS trigger, and Firebase Realtime Database for sub-200ms GPS-sharing with pre-configured contacts.

Gestura — Bidirectional Sign-Language AI — *Python, MediaPipe, OpenAI Whisper, PyTorch, Stable Diffusion*

- Built a real-time sign-to-speech pipeline: MediaPipe hand-key point tracking feeding a custom-trained MLP classifier, with spoken output via gTTS.
- Built the reverse pipeline — OpenAI Whisper speech transcription driving a Stable Diffusion + ControlNet avatar generator to render sign-language poses.

IoT Environmental Monitoring Suite — *ESP32, Django, ThingSpeak, MQTT*

- Built an air-quality monitor (ESP32 + MQ-135) tracking CO2, NH3 and benzene levels, streaming readings to ThingSpeak with a Django + Bootstrap dashboard.
- Built a smart irrigation system (ESP32 + DHT11 + soil-moisture sensors) automating watering and logging crop-health data via MQTT.

CERTIFICATIONS & ACHIEVEMENTS

- Python Programming – GUVI (HCL) | Java Training – LinkedIn Learning | SQL Bootcamp – Lets Upgrade
- Deloitte & TATA Cybersecurity Simulations (Forge) — threat analysis, SIEM triage, IAM, vulnerability assessment
- MeitY–ISEA Digital Security Quiz — Ministry of Electronics & IT, Government of India